

CHE 201 Spring 2020 Quiz 1

Anna Christensen

TOTAL POINTS

100 / 100

QUESTION 1

1 Question 1 60 / 60

✓ - **0 pts** Correct

- **10 pts** Patm is wrong
- **20 pts** formulas are wrong
- **5 pts** unit is wrong
- **5 pts** mistake with the formula
- **60 pts** not done
- **10 pts** calculation error

QUESTION 2

2 Question 2 40 / 40

✓ - **0 pts** Correct

- **5 pts** units are missing
- **5 pts** system boundary not labeled
- **40 pts** not done
- **5 pts** P and V values not marked
- **10 pts** state 2 is wrong

NAME: Anna ChristensenUIN: 670565580**QUIZ 1: 2 problems (10 pts)****Closed book/Closed notes: Show all your work for full credit**

- (6 pts) The average atmospheric pressure on Earth is approximated as a function of altitude by the relation $P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256}$, where P_{atm} is atmospheric pressure in kPa and z is altitude in km with $z=0$ at sea level. A monometer is used to measure the pressure of a gas in a tank in Denver ($z=1610\text{m}$). The fluid has a density of 850kg/m^3 and the monometer column height is 200cm . Determine the absolute and gage pressure of gas within the tank.
- (4 pts) A gas contained within a piston-cylinder assembly undergoes two processes in series:

Process 1-2: Constant-pressure compression at 1 bar from $V_1 = 2.5\text{ m}^3$ to $V_2 = 1\text{ m}^3$ **Process 2-3:** Constant volume compression from 1 bar to 2 bar

- Draw a schematic of the system and label the system boundary.
- Sketch the processes on a p-V diagram; label States 1, 2 and 3 and the pressure and volume values at each numbered state.

(2) process 1-2

$$P_1 = 1\text{ bar} \quad P_2 = 1\text{ bar}$$

$$V_1 = 2.5\text{ m}^3 \quad V_2 = 1\text{ m}^3$$

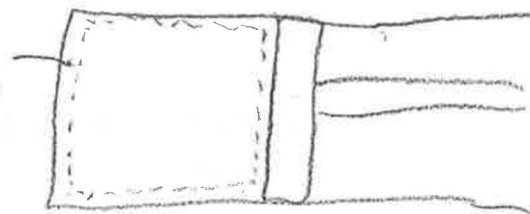
process 2-3

$$P_2 = 1\text{ bar} \quad P_3 = 2\text{ bar}$$

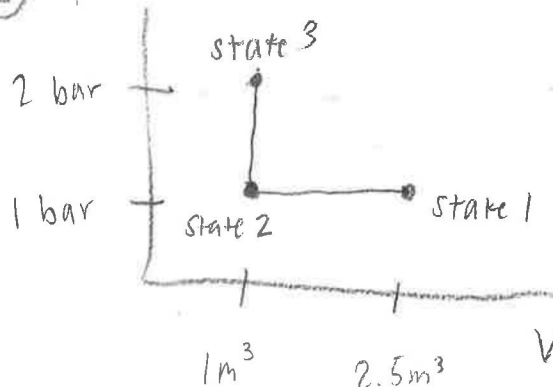
$$V_2 = 1\text{ m}^3 \quad V_3 = 1\text{ m}^3$$

(a)

system boundary



(b) P



$$\textcircled{1} \quad P_{\text{atm}} = 101.325 (1 - 0.02256 Z)^{5.256} \text{ kPa}$$

$$Z = 1610 \text{ m} = 1.61 \text{ km}$$

$$P_{\text{atm}} = 83.42 \text{ kPa} = 83420 \text{ Pa}$$

DENVER

$$P_{\text{gage}} = P_{\text{abs}} - P_{\text{atm}}$$

$$\rho = 850 \frac{\text{kg}}{\text{m}^3}$$

$$h = 200 \text{ cm} = 2 \text{ m}$$

$$P_{\text{gage}} = \rho g h + P_{\text{atm}} - P_{\text{atm}}$$

$$P_{\text{abs}} = \rho g h + P_{\text{atm}}$$

$$P_{\text{gage}} = \rho g h$$

$$P_{\text{abs}} = 850 \frac{\text{kg}}{\text{m}^3} \left(9.8 \frac{\text{m}}{\text{s}^2} \right) (2 \text{ m}) + 83420 \text{ Pa}$$

$$P_{\text{abs}} = 100080 \text{ Pa} = 100.08 \text{ kPa}$$

$$P_{\text{abs}} = 100.08 \text{ kPa}$$

$$P_{\text{gage}} = 100080 \text{ Pa} - 83420 \text{ Pa} = 16660 \text{ Pa} = \rho g h$$

$$P_{\text{gage}} = 16.66 \text{ kPa}$$

